A four-moment proof of the conjectured recurrence for A172671

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Abstract

Let a_n count the $3n \times 6$ matrices with entries in $\{0, 1, 2\}$, row sums 2, and column sums n, with $a_0 = 1$. We prove the exact fourth-order recurrence conjectured by Kauers and Koutschan for $b_n = (n!)^3 a_n/(3n)!$, for every integer $n \geq 0$. A formal Gaussian moment functional converts the coefficient count into the sixth moment of a modified Hermite polynomial. Integration by parts closes on four moments and gives an explicit 4×4 transfer matrix. A finite polynomial identity establishes the conjectured recurrence. Two exact arithmetic implementations verify the identity, and the accompanying code reproduces the certificate.

1 Counting problem, prior work, and exact result

Rows and columns are labeled. A row of sum 2 is either 2 in one position, or 1 in two distinct positions. There are $6 + 15 = 21$ choices. Consequently, with

$$h(\boldsymbol{t}) = \sum_{1 \leq i \leq j \leq 6} t_i t_j = \frac{1}{2} \left(\left(\sum_{i=1}^{6} t_i \right)^2 + \sum_{i=1}^{6} t_i^2 \right),$$

the exact counting formula is

$$a_n = [t_1^n \cdots t_6^n] h(\boldsymbol{t})^{3n}, \qquad b_n = \frac{(n!)^3}{(3n)!} a_n. \tag{1}$$

The convention $a_0 = 1$ counts the empty matrix. The OEIS entry A172671 [2] starts at $n = 1$; we explicitly include $n = 0$.

Kauers and Koutschan [1, Section 4.2, Conjecture 7] give the recurrence below, and their Theorem 8 proves D-finiteness without establishing this particular recurrence. Their final JIS article labels it a conjecture on page 20. Yurkevich [3, Conjecture 10.60 and equation (10.5), pp. 180–181] also states the conjecture and uses the displayed decomposition of h to compute terms more efficiently. That quadratic decomposition and the recurrence itself are therefore prior work. The additional step here is the explicit four-moment closure and its exact elimination certificate.

Theorem 1. *For every integer $n \geq 0$, the sequence defined by (1) satisfies*

$$\sum_{j=0}^{4} p_j(n) b_{n+j} = 0, \tag{2}$$

where the coefficients are exactly

$$\begin{aligned}
p_4(n) &= 3(n+3)(n+4)^3(3784n^4 + 32164n^3 + 100749n^2 + 137862n + 69678),\\
p_3(n) &= -(n+3)(3799136n^7 + 72183584n^6 + 579689880n^5\\
&\quad + 2548427912n^4 + 6617561702n^3 + 10141503096n^2\\
&\quad + 8487349821n + 2991586122),\\
p_2(n) &= -3(10844944n^8 + 222321352n^7 + 1973930222n^6\\
&\quad + 9916013134n^5 + 30831383530n^4 + 60768378830n^3\\
&\quad + 74160044251n^2 + 51243135187n + 15352797306),\\
p_1(n) &= (n+2)(29681696n^7 + 504588832n^6 + 3602458816n^5\\
&\quad + 14001842392n^4 + 32010306742n^3 + 43078657918n^2\\
&\quad + 31639900193n + 9799573455),\\
p_0(n) &= 15435(n+1)^3(n+2)(3784n^4 + 47300n^3 + 219945n^2\\
&\quad + 450988n + 344237).
\end{aligned}$$

The initial conditions are

$$b_0 = 1, \qquad b_1 = 15, \qquad b_2 = 2249, \qquad b_3 = 444975. \tag{3}$$

Since $p_4(n) > 0$ for $n \geq 0$, these four values and (2) determine the entire sequence.

2 An exact Gaussian-moment representation

Define a linear functional $\mathcal{L} : \mathbb{Q}[x] \to \mathbb{Q}$ by

$$\mathcal{L}(x^{2m}) = (2m-1)!! = \frac{(2m)!}{2^m m!}, \qquad \mathcal{L}(x^{2m+1}) = 0, \qquad (-1)!! = 1. \tag{4}$$

It is the standard normal expectation on polynomials, but the proof uses only the following algebraic identities:

$$\mathcal{L}(f') = \mathcal{L}(xf), \qquad \mathcal{L}(e^{tx}) = e^{t^2/2}. \tag{5}$$

For the first identity it suffices to take $f = x^r$: both sides vanish when r is even, while for $r = 2m+1$ they are both $(2m+1)!!$. The second identity follows coefficient by coefficient from (4). Exponential series below are formal series, and $\mathcal{L}$ is applied to their polynomial coefficients. No convergence or interchange of infinite analytic integrals is required.

Define the modified Hermite polynomials by

$$e^{tx+t^2/2} = \sum_{n\geq 0} P_n(x)\frac{t^n}{n!}, \qquad P_n(x) = n! \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{x^{n-2k}}{2^k k!(n-2k)!}. \tag{6}$$

The first polynomials are $1, x, x^2+1, x^3+3x$. Differentiation of the generating series gives, for all $n \geq 0$,

$$P_{n+1} = xP_n + P_n', \qquad P_n'' = nP_n - xP_n', \qquad P_{n+1}' = (n+1)P_n. \tag{7}$$

For example, $\partial_x e^{tx+t^2/2} = te^{tx+t^2/2}$ gives $P_n' = nP_{n-1}$ for $n \geq 1$, while $\partial_t e^{tx+t^2/2} = (x+t)e^{tx+t^2/2}$ gives the first relation; differentiating it gives the remaining identities. At $n = 0$ they hold directly since $P_0 = 1$.

Lemma 2. *For every $n \geq 0$, $b_n = \mathcal{L}(P_n^6)/(n!)^3$.*

Proof. By (5), in the ring of formal series in $t_1, \ldots, t_6$,

$$\mathcal{L}\left(\prod_{i=1}^{6} e^{xt_i + t_i^2/2}\right) = \exp\left(\frac{1}{2}\Big(\sum_i t_i\Big)^2 + \frac{1}{2}\sum_i t_i^2\right) = e^{h(\boldsymbol{t})}.$$

The coefficient of $t_1^n \cdots t_6^n$ on the left is $\mathcal{L}(P_n^6)/(n!)^6$. Since h is homogeneous of degree 2, only $h^{3n}/(3n)!$ contributes to that coefficient on the right, which is $a_n/(3n)!$. Multiplying by $(n!)^3$ proves the claim. □

3 Closure on four moments

Fix $n \geq 0$, and put $y = P_n(x)$ and $z = P_n'(x)$. Then $y' = z$ and $z' = ny - xz$. For $0 \leq j \leq 6$ write

$$e_j = y^{6-j} z^j, \qquad M_{k,j} = \mathcal{L}(x^k e_j) \quad (k \geq 0).$$

Differentiating $x^k e_j$ and using (5) gives

$$(j+1)M_{k+1,j} = kM_{k-1,j} + (6-j)M_{k,j+1} + jnM_{k,j-1}. \tag{8}$$

The first term is omitted at $k = 0$. At $j = 0$ and $j = 6$, respectively, the out-of-range term is absent because its coefficient is zero. Explicitly,

$$(x^k e_j)' = kx^{k-1}e_j + (6-j)x^k e_{j+1} + jnx^k e_{j-1} - jx^{k+1}e_j,$$

so the factor $j+1$ in (8) includes the contribution from the Gaussian weight. In an integral formulation the boundary term is a polynomial times $e^{-x^2/2}$, hence vanishes at both infinities; the algebraic proof already accounts for it exactly.

By (6), $P_n(-x) = (-1)^n P_n(x)$, so e_j has parity j. In particular $M_{0,j} = 0$ for odd j. Induction on k in (8) expresses every $M_{k,j}$ as a $\mathbb{Q}[n]$-linear combination of the four entries of

$$W_n = \big(\mathcal{L}(y^6), \mathcal{L}(y^4 z^2), \mathcal{L}(y^2 z^4), \mathcal{L}(z^6)\big)^{\mathsf{T}}.$$

All divisions in this induction are by the positive integers $j+1$; there are no exceptional values of n.

Lemma 3. *For every integer $n \geq 0$,*

$$W_{n+1} = A(n)W_n, \qquad U_{n+1} = B(n)U_n, \quad U_n = \frac{W_n}{(n!)^3}, \quad B(n) = \frac{A(n)}{(n+1)^3}, \tag{9}$$

where

$$A(n) = \begin{pmatrix} 3(2n+1)(29n^2+29n+5) & 15(76n^2+95n+28) & 75(4n+3) & 7 \\ \dfrac{(n+1)^2(76n^2+57n+9)}{3} & \dfrac{241(n+1)^2(2n+1)}{3} & 35(n+1)^2 & 0 \\ (n+1)^4(4n+1) & 21(n+1)^4 & 0 & 0 \\ (n+1)^6 & 0 & 0 & 0 \end{pmatrix}. \tag{10}$$

The first component of U_n is b_n, and $U_0 = (1,0,0,0)^{\mathsf{T}}$.

Proof. Under $n \mapsto n+1$, (7) replaces (y, z) by $(xy + z, (n+1)y)$. Thus, for $j = 0, 2, 4, 6$,

$$\mathcal{L}(P_{n+1}^{6-j}(P'_{n+1})^j) = (n+1)^j \sum_{\ell=0}^{6-j} \binom{6-j}{\ell} M_{6-j-\ell,\ell}. \tag{11}$$

Applying (8) produces the corresponding row of (10). For completeness, an exact finite specification of this computation is as follows. Let $R_{k,j}$ be a row of length four, set $R_{0,2r}$ equal to the rth unit row ($r = 0, 1, 2, 3$), and set $R_{0,j} = 0$ for odd j. Set out-of-range rows to zero. For $k \geq 1$,

$$R_{k,j} = \frac{(k-1)R_{k-2,j} + (6-j)R_{k-1,j+1} + jnR_{k-1,j-1}}{j+1}, \tag{12}$$

omitting the first term when $k = 1$. Row r of $A(n)$ is then

$$(n+1)^{2r} \sum_{\ell=0}^{6-2r} \binom{6-2r}{\ell} R_{6-2r-\ell,\ell}. \tag{13}$$

Equations (12)–(13) verify all sixteen displayed entries using only $k \leq 6$. For instance, $M_{2,0} = M_{0,0} + 6M_{1,1}$ and $M_{1,1} = (5M_{0,2} + nM_{0,0})/2$. Hence the row for $j = 4$ is $(n+1)^4((4n+1)M_{0,0} + 21M_{0,2})$, as displayed. Normalization gives $B(n)$. The final assertions follow from Lemma 2 and $P_0 = 1$, $P'_0 = 0$. □

4 An exact elimination certificate

Let $\epsilon = (1, 0, 0, 0)$, and define matrix polynomials

$$C_0(n) = I_4, \qquad C_j(n) = A(n+j-1)C_{j-1}(n) \quad (1 \leq j \leq 4).$$

The order of multiplication matters: explicitly, $C_j(n) = A(n+j-1)\cdots A(n)$.

Lemma 4. *With the five polynomials in Theorem 1, the following is an identity of polynomial rows over* $\mathbb{Q}[n]$*:*

$$\boxed{\sum_{j=0}^{4} p_j(n) \left(\prod_{t=j+1}^{4} (n+t)^3 \right) \epsilon C_j(n) = (0, 0, 0, 0).} \tag{14}$$

An empty product is 1.

Proof. This is a finite polynomial identity with all of its inputs explicitly given in (10) and Theorem 1. It can be verified by polynomial multiplication and coefficient collection as follows. Starting with $C = I_4$ and $H = (0, 0, 0, 0)$, for $j = 0, \ldots, 4$ replace C by $A(n+j-1)C$ if $j > 0$, and add $p_j(n) \prod_{t=j+1}^{4} (n+t)^3$ times the first row of C to H. The resulting polynomial coefficients are all zero. The sizes of the four coordinate checks are given in Table 1; these are identities in the indeterminate n, not evaluations at sample indices.

Coordinate	0	1	2	3
Maximum degree of any summand	20	19	18	17
Coefficients checked (including constant)	21	20	19	18
Nonzero coefficients after summation	0	0	0	0

Table 1: Exact rational-polynomial certificate: all 78 coefficients vanish.

The calculation has been checked with two separate exact-arithmetic programs. One uses SymPy rational functions; the other implements polynomial addition, multiplication, substitution

$n \mapsto n + h$, and coefficient collection from scratch using rational numbers. The latter also reconstructs (10) from (12)–(13) and reads the five target polynomials directly from the original TeX excerpt. Thus neither recurrence guessing nor numerical tolerance enters the certificate. Appendix A gives precise coefficient rules sufficient to reproduce the calculation using rational arithmetic alone. □

Proof of Theorem 1. Iterating (9) gives

$$b_{n+j} = \frac{\epsilon C_j(n) U_n}{\prod_{t=1}^{j}(n+t)^3} \qquad (0 \le j \le 4).$$

For $n \ge 0$, the common denominator $D(n) = \prod_{t=1}^{4}(n+t)^3$ is strictly positive. Multiply the desired recurrence by $D(n)$, substitute the last display, and apply (14) to U_n. This proves the exact recurrence on the full claimed domain, including $n = 0$.

The initial values can be computed without using that recurrence:

$$\mathcal{L}(1) = 1, \quad \mathcal{L}(x^6) = 15, \quad \mathcal{L}((x^2+1)^6) = 17992, \quad \mathcal{L}((x^3+3x)^6) = 96114600.$$

Division by $(n!)^3$, for $n = 0, 1, 2, 3$, gives (3). Every factor of $p_4(n)$ outside its quartic is positive, and all coefficients of that quartic are positive. Thus no singular starting index occurs for $n \ge 0$, and forward induction proves the uniqueness assertion. □

5 Independent checks and reproducibility

The proof above uses exact algebra for unbounded n. Separately, a direct coefficient dynamic program checks the original matrix interpretation. If $F_r(\boldsymbol{c})$ counts ordered lists of r admissible rows with column sums $\boldsymbol{c}$, then

$$F_0(\mathbf{0}) = 1, \qquad F_{r+1}(\boldsymbol{c}) = \sum_{\boldsymbol{s} \in S} F_r(\boldsymbol{c} - \boldsymbol{s}), \qquad a_n = F_{3n}(n, n, n, n, n, n),$$

where S consists of the 21 row choices and any negative coordinate gives zero. This independently agrees with (6) and Lemma 2 for $0 \le n \le 8$. Further exact sixth-moment computations through $n = 40$ give zero recurrence residual for $0 \le n \le 36$. These finite tests corroborate the proof; they are not its justification.

n	a_n	b_n
0	1	1
1	90	15
2	202410	2249
3	747558000	444975
4	3536978063850	102077289
5	19292117692187340	25493181015

Table 2: Exact examples, with the empty-matrix convention at $n = 0$.

The accompanying package contains the source excerpt, symbolic derivation, independent rational-polynomial verifier, direct-count checks, and their JSON reports. From the reproducibility directory the following commands reproduce the checks:

```
python src/verify_certificate.py
python src/route_alt_transfer.py
python src/independent_checks.py
```

The first command needs only the Python standard library. The second uses SymPy; the third uses the standard library. Paths in the scripts are resolved relative to their own locations. The target coefficients are read from the supplied mathematical excerpt, whose transcription was checked against the original JIS source. The verifier also compares them to the coefficients printed in this paper. No network access is required. The paper can be rebuilt with two runs of

```
pdflatex -interaction=nonstopmode -halt-on-error -output-directory=output paper/main.tex
```

Rename the resulting `output/main.pdf` to `output/research_paper.pdf`, or use the supplied `src/build_paper.ps1` helper. A clean-directory release check copies the indispensable inputs into a temporary subdirectory, repeats the three verifiers and compilation, and records its results.

Authorship and verification. GPT-6 Astra is an AI system. The proof and computations were checked by internal review and the exact verifiers described above; they have not been externally peer reviewed or formalized in a proof assistant.

A Coefficient arithmetic for the certificate

Here is a complete finite arithmetic specification of the identity check. Represent a polynomial f by its list of rational coefficients $[f_0, \dots, f_d]$, with omitted coefficients interpreted as zero. Use

$$[n^k](f+g) = f_k + g_k,$$
$$[n^k](fg) = \sum_{i=0}^{k} f_i g_{k-i},$$
$$[n^k]f(n+h) = \sum_{i=k}^{d} \binom{i}{k} h^{i-k} f_i.$$

Apply addition and multiplication entrywise in the usual matrix-product rule. Expand the factored entries in (10) and the five polynomials of Theorem 1 by these rules, retaining exact rational numbers throughout. The following pseudocode fixes all loop bounds and the multiplication order:

```
C := identity_matrix(4)
H := zero_row(4)
for j = 0,1,2,3,4:
    if j > 0:
        C := shift(A, j-1) * C
    q := p[j]
    for t = j+1,...,4:
        q := q * (n+t)^3
    H := H + q * first_row(C)
assert H[0] == 0 and H[1] == 0 and H[2] == 0 and H[3] == 0
```

In the final line equality means equality of every rational coefficient. No interpolation, inferred annihilator, floating-point arithmetic, or unproved closure assertion is used. Together with the displayed polynomials and matrix, this algorithm fully specifies the 78 exact zero-coefficient checks of Lemma 4.

References

[1] M. Kauers and C. Koutschan, *Some D-finite and some possibly D-finite sequences in the OEIS*, Journal of Integer Sequences **26** (2023), Article 23.4.5, Section 4.2, Conjecture 7 and Theorem 8. https://cs.uwaterloo.ca/journals/JIS/VOL26/Koutschan/kout4.pdf. Preprint: https://arxiv.org/abs/2303.02793v2, 24 April 2023.

[2] The OEIS Foundation Inc., *Entry A172671*, https://oeis.org/A172671, accessed 30 October 2026.

[3] S. Yurkevich, *Integer sequences, algebraic series and differential operators*, doctoral dissertation, University of Vienna, 2023, Conjecture 10.60 and equation (10.5), pp. 180–181. https://homepage.univie.ac.at/herwig.hauser/Publications/yurkevich%20dissertation%202023.pdf.